

Project Design Phase Problem – Solution Fit Template

Date	15 February 2025
Team ID	
Project Name	AI FAQ Assistant API
Maximum Marks	2 Marks

Problem – Solution Fit Template:

Problem-Solution Fit means you have found a real problem for your customer and the solution you have designed actually solves it. For AI FAQ Assistant API it helps the team understand how support teams, customers and developers behave, and what will work and why.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase adoption by tapping into existing channels of behaviour.
- Sharpen communication and marketing with the right triggers and messaging.
- Increase touch-points by solving frequent, urgent or costly problems and building trust.
- Understand the existing situation in order to improve it for the target group.

References:

<https://www.ideahackers.network/problem-solution-fit-canvas/>

<https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

Template:

1. CUSTOMER SEGMENT(S) [CS] <i>Who is your customer?</i> • Small and mid-size businesses with online customers • Support managers and customer care teams • Developers and agencies building websites, apps and helpdesks • SaaS startups with product FAQs	6. CUSTOMER CONSTRAINTS [CC] <i>What prevents customers from taking action?</i> • Small support team and tight budget • Fear of wrong or invented AI answers • Customer data and privacy concerns • Little time or skill to build and tune AI search	5. AVAILABLE SOLUTIONS [AS] <i>Which solutions exist today? Pros and cons</i> • Static FAQ pages: free, but exact-word search and rarely read • Generic chatbots: quick start, but not tied to company content and may invent answers • Full helpdesk suites with AI: powerful, but costly and heavy • Email and phone: accurate, but slow and hard to scale
2. JOBS-TO-BE-DONE / PROBLEMS [J&P] <i>Which problems do you address?</i> • Answer common questions instantly, at any hour • Keep answers accurate and based on approved FAQs • Hand over to a human when unsure • Keep FAQs up to date from real questions • Offer all of this through one simple API	9. PROBLEM ROOT CAUSE [RC] <i>What is the real reason this problem exists?</i> • Customers phrase questions differently from FAQ text, and keyword search misses the meaning • Chatbots are not grounded in company content • Question volume grows faster than the support team • Teams must build retrieval, security and fallback themselves	7. BEHAVIOUR [BE] <i>What does the customer do to get the job done?</i> • Customers skim the FAQ page, then email or chat anyway • Agents copy and paste the same replies • Businesses try chatbot widgets and drop those that give wrong answers • Developers wire LLM APIs to their own FAQ database
3. TRIGGERS [TR] <i>What triggers customers to act?</i> • A jump in support tickets after a launch or sale • Complaints about slow replies • Pressure to add a help assistant to the site quickly	10. YOUR SOLUTION [SL] <i>A solution that fits customer limits, solves the problem and matches behaviour.</i> • AI FAQ Assistant API: a NestJS REST backend that answers questions from a company's own FAQs • Semantic search (embeddings) finds the right FAQ • AI writes a short answer with sources and a confidence score • Low confidence means a clear "I don't know" and human handoff • Feedback and analytics reveal missing FAQs • API keys, rate limiting, Swagger docs and a free tier	8. CHANNELS of BEHAVIOUR [CH] 8.1 ONLINE • Searches for AI chatbot and FAQ API tools • Developer communities (GitHub, Stack Overflow, Dev.to) • Product directories and review sites
4. EMOTIONS: BEFORE / AFTER [EM] <i>How do customers feel before and after?</i> • Before: overwhelmed, worried about complaints • After: relieved, confident, in control		8.2 OFFLINE • Word of mouth between business owners • Meetups, workshops and hackathons • Agency and consultant recommendations